

Image registration for microscopy images

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Outline

- What is image registration?
- Methods in image registration
- Types of transformations
- Available tools
- Demonstration

Image registration

- Image registration also termed as image alignment is the process of aligning (overlying) different images over each other
- It could also be done on different sets of image data to bring to one coordinate system
- It is necessary in order to be able to compare or integrate the data obtained from different measurements

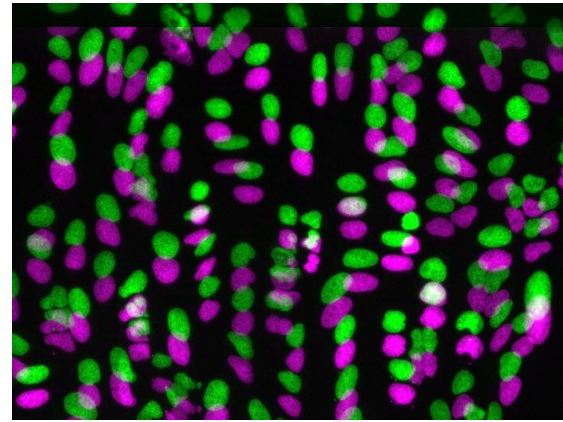
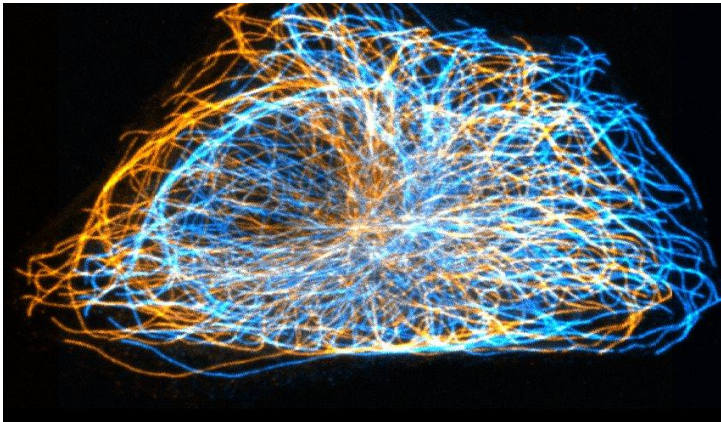
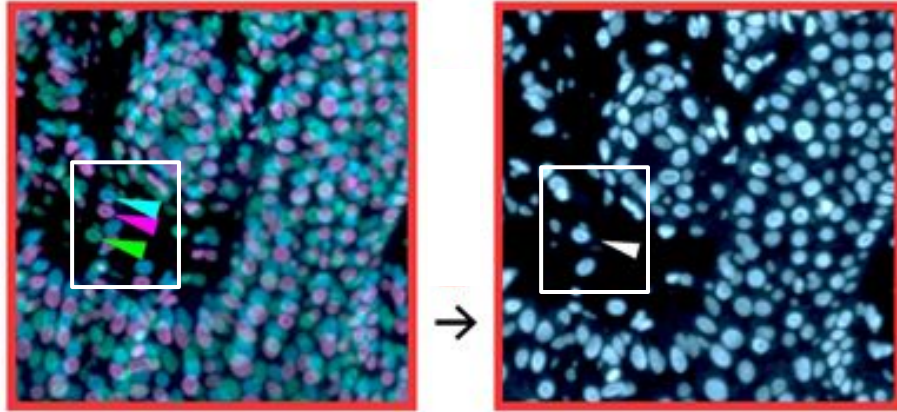


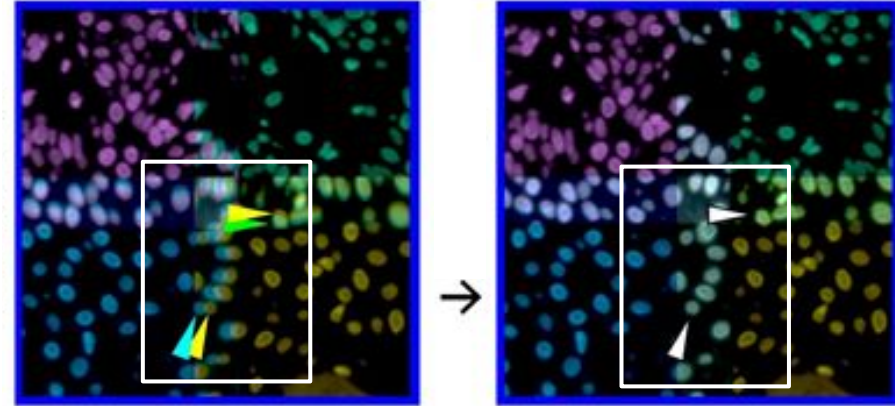
Image from BBBC (Gustafsdottir *et al.*, 2013)

Scenarios

Cyclic imaging

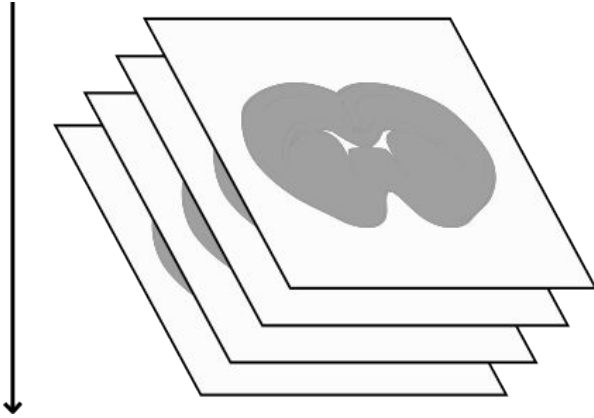


Tiled images

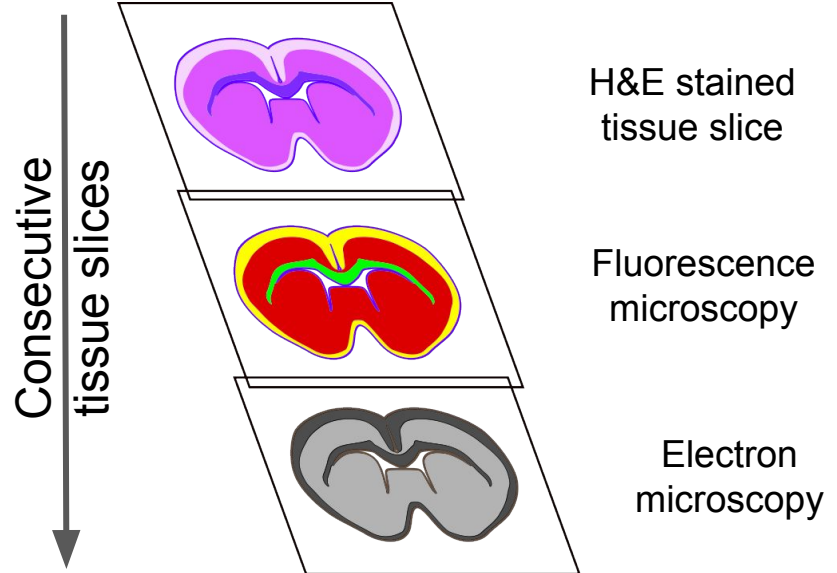


Scenarios

Serial sections



Multimodal images



Basic terminology

Input image / Moving image - the image that will be transformed

Reference image / Fixed image - is the image against which the input image is registered

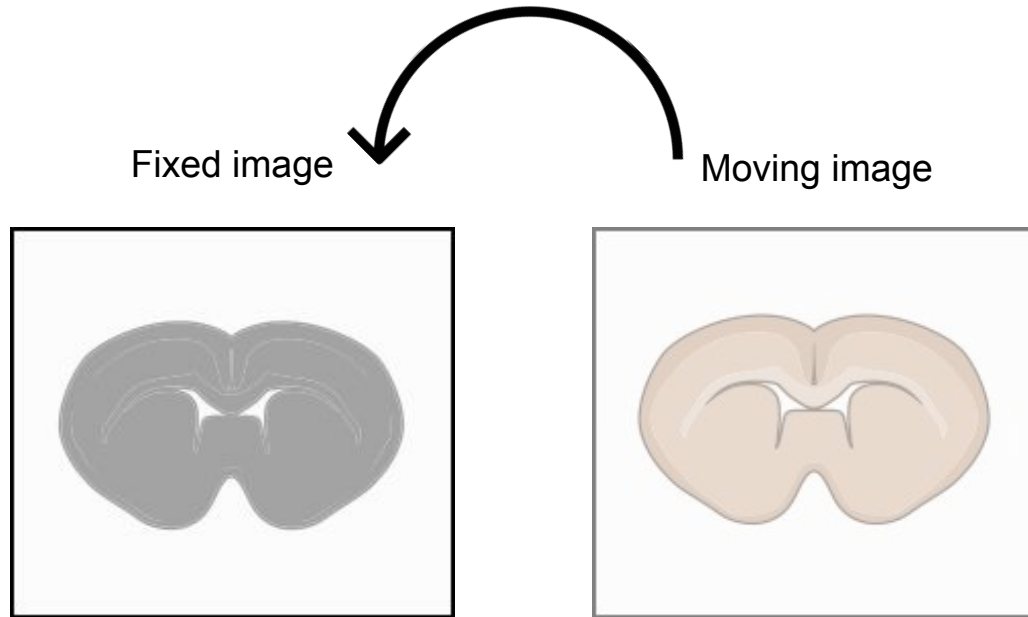
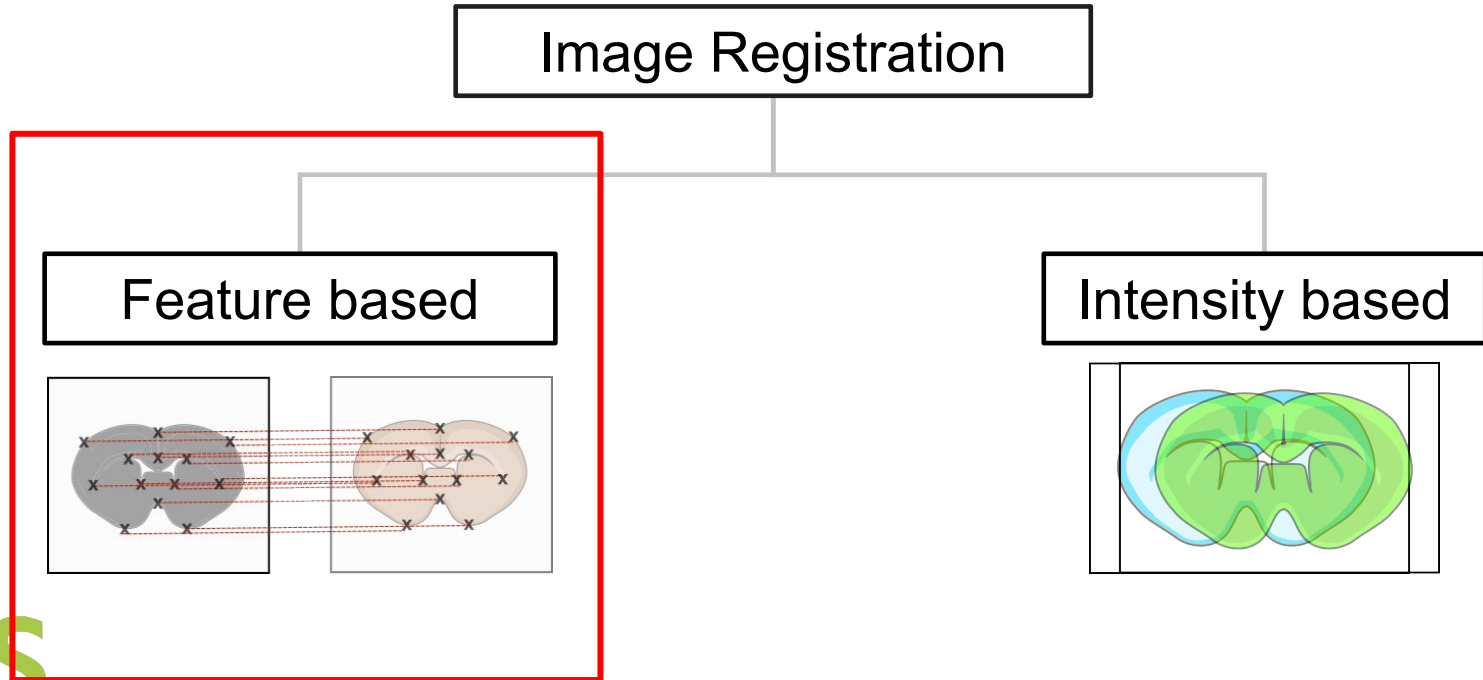


Image registration - Methods



Image registration - Methods



Feature-based image registration

Feature detection

Feature matching

Transform model estimation

Image transformation

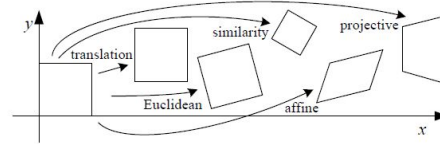
Fixed image

Moving image

Fixed image

Moving image

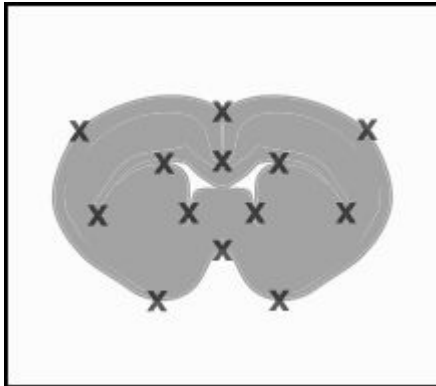
Registered image



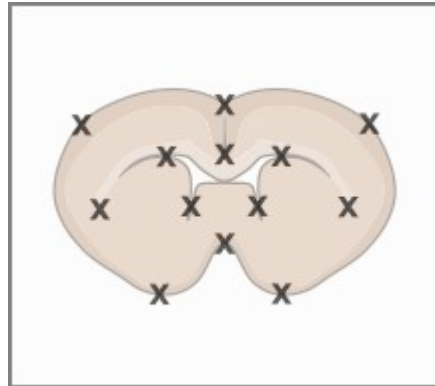
Feature detection

- Salient and distinctive objects that are common in both the images are considered
- Point features, line features, edges, region features are some examples
- These are called ***control points/ keypoints / correspondence points***
- The detected points should be spread over the images, stable in time and not be sensitive to illumination changes.

Fixed image

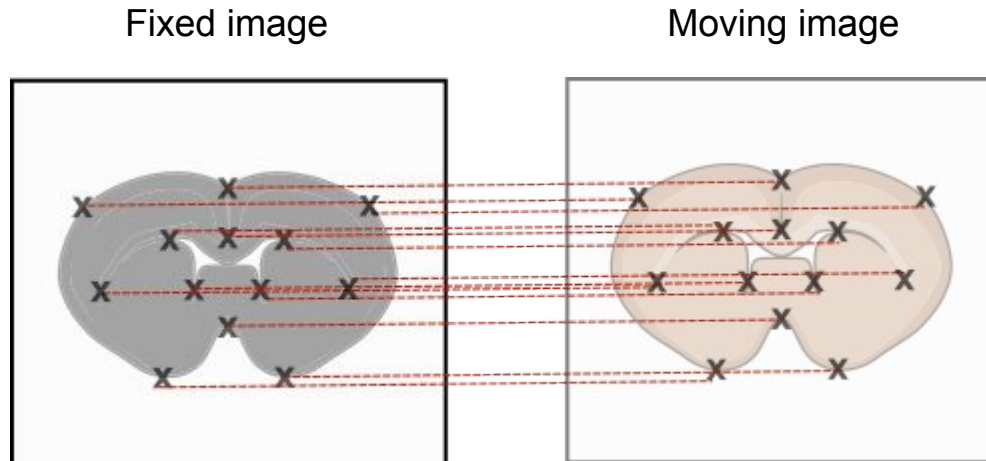


Moving image



Feature matching

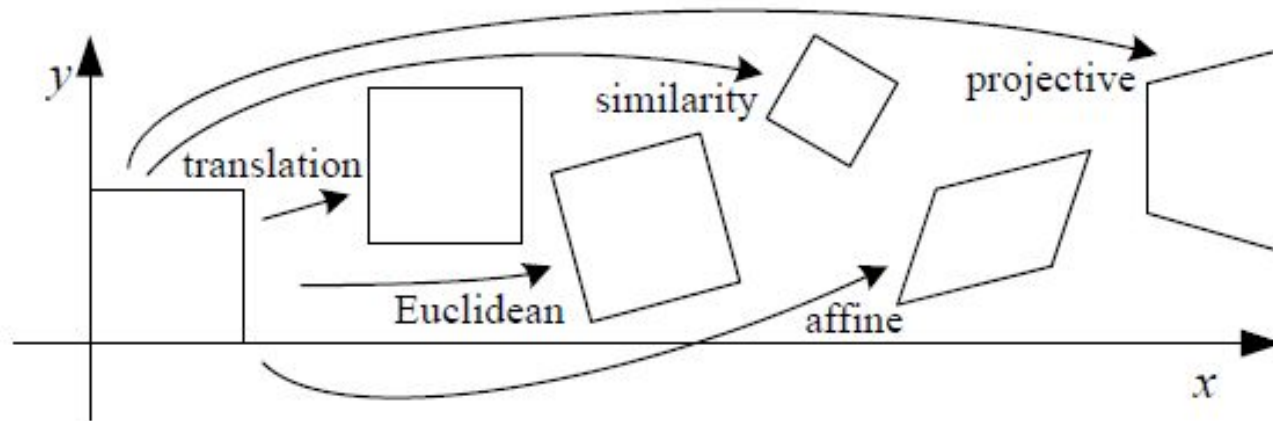
- The correspondence between the features detected in the input image and those detected in the reference image is established.
- Feature descriptors and similarity measures along with spatial relationships among the features are used for matching.



Transform model estimation - Methods

- Linear / non-deformable transformation
- Non-linear / deformable transformation

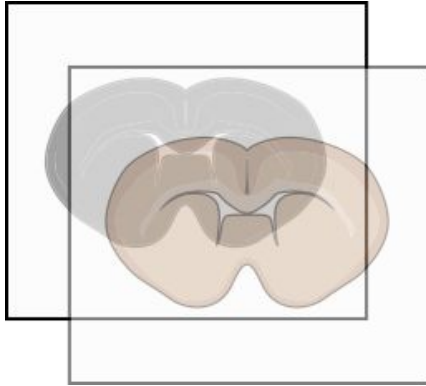
Linear transformations



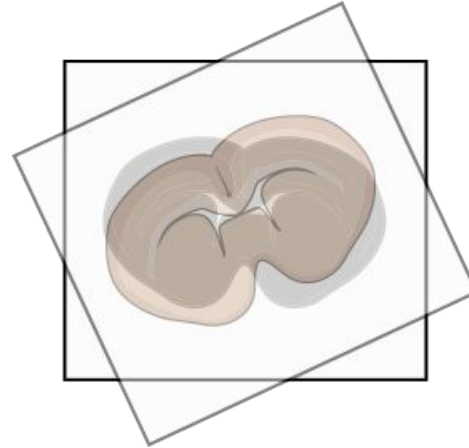
Linear transformations

Rigid/Euclidean transformation

Translation



Rotation

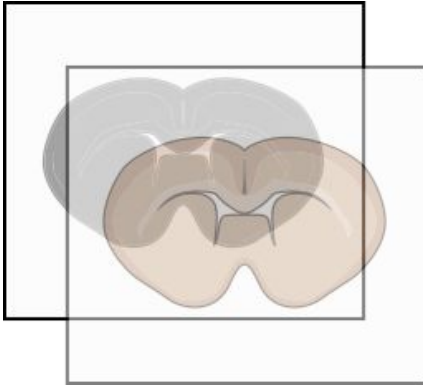


Translation - parameters - 2 (x, y) ; preserves orientation
Rigid transformation - parameters - 3 (x, y, θ); preserves lengths

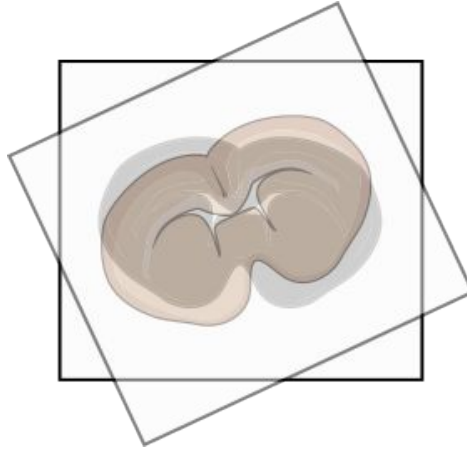
Linear transformations

Similarity transformation

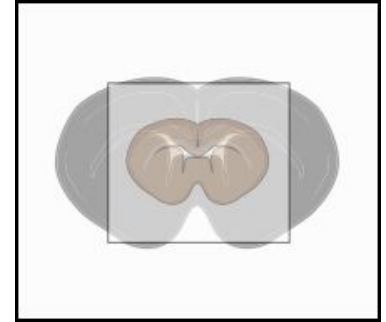
Translation



Rotation



Scaling

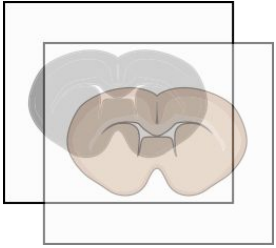


***Similarity transformation - parameters - 4;
preserves angles, parallelism and straight lines***

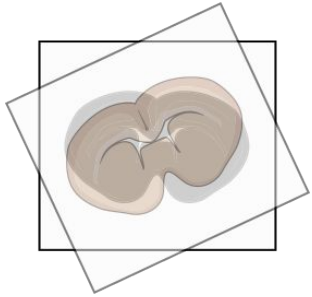
Linear transformations

Affine transformation

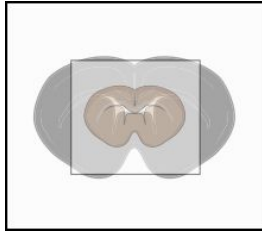
Translation



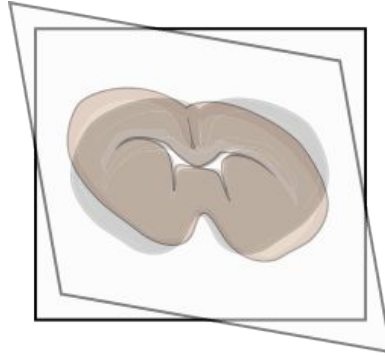
Rotation



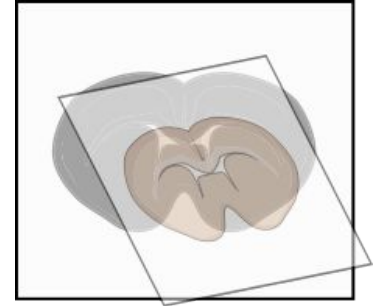
Scaling



Shear

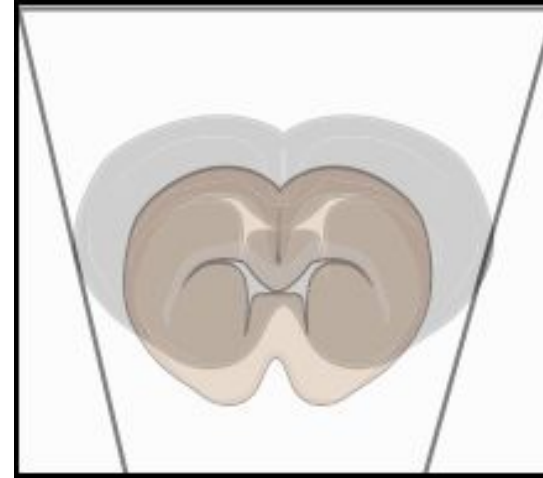
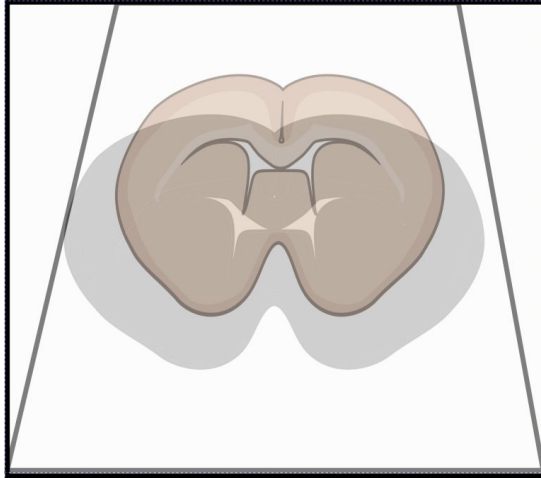


Example of Affine



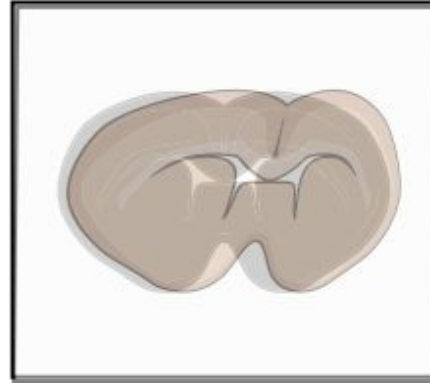
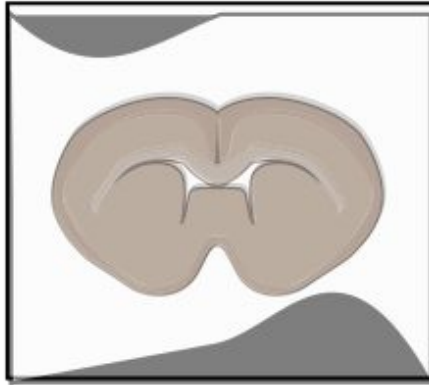
Linear transformations

Projective transformation



Non-linear transformations

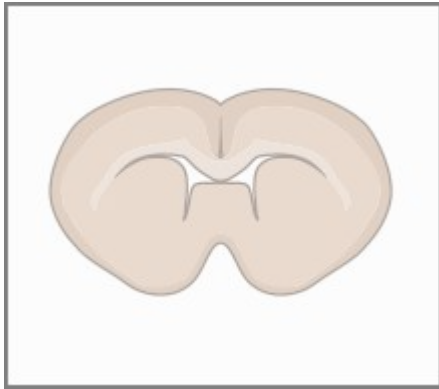
Deformable transformation



How do the transformations take place?

Linear transformations

Moving image



$\times$

Scaling & Rotation

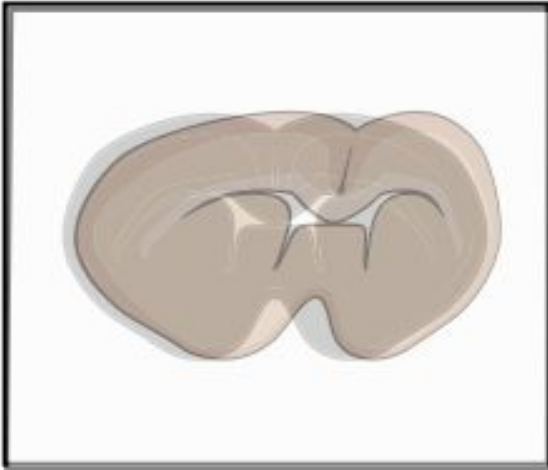
$$\begin{Bmatrix} a_{11} & a_{12} & t_1 \\ a_{21} & a_{22} & t_2 \\ 0 & 0 & 1 \end{Bmatrix}$$

Translation

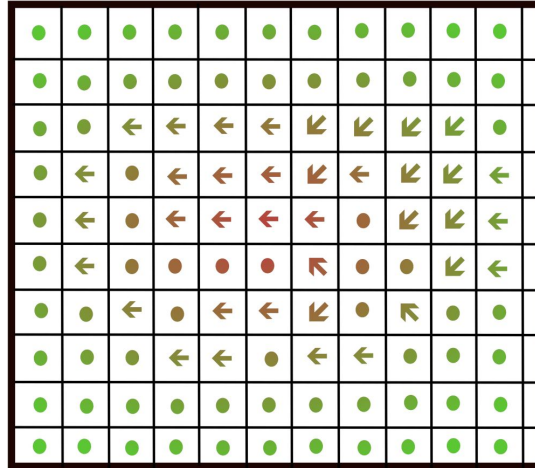
How do the transformations take place?

Non-linear transformations

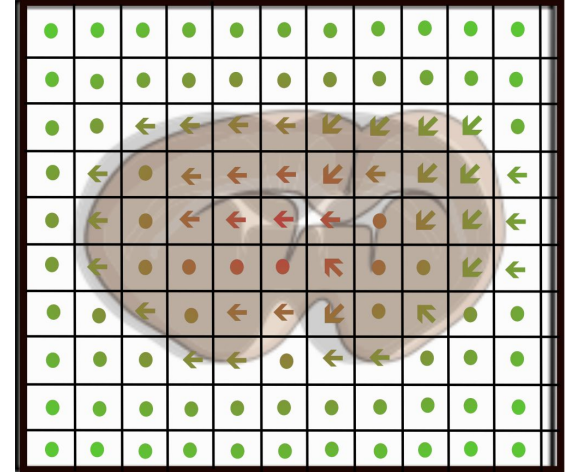
Sample image



Displacement



Overlay



Evaluation

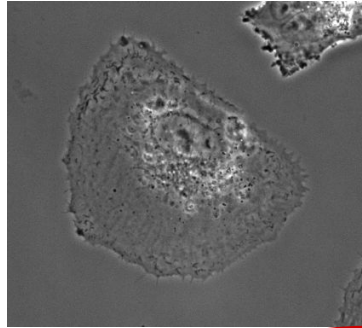
Pearson correlation coefficient - higher the better

Sum of squared differences - lower the better

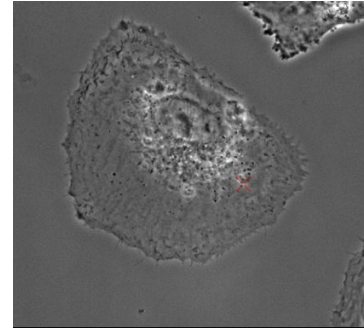
*Feature-based image registration methods mostly use ***L2 norm*** or ***Hamming distance*** as evaluation criteria

Example

Frame 1



Frame 2

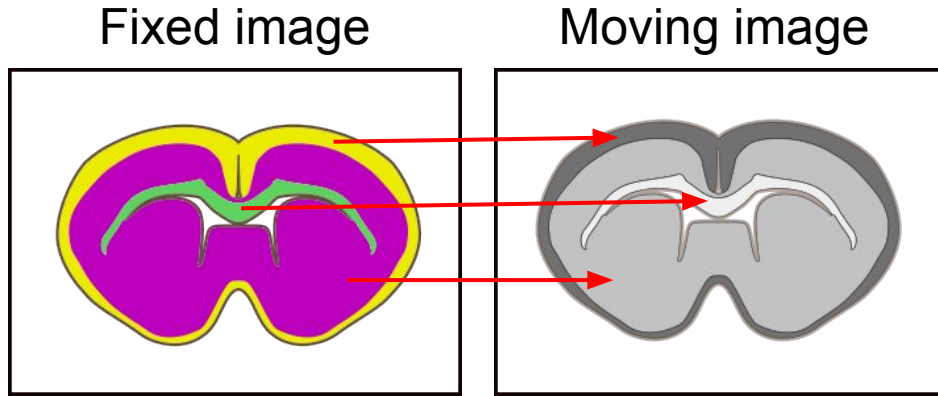


Pixel values	Prefs	137	138	139	140	141	142	143
	63	103	105	97	97	97	95	103
64	95	103	99	109	104	103	108	
65	78	84	98	110	101	110	104	
66	77	80	95	113	113	105	109	
67	70	66	86	109	111	108	105	
68	65	71	75	102	116	117	121	
69	59	72	79	103	127	138	133	

Pixel values	Prefs	137	138	139	140	141	142	143
	63	93	97	106	91	71	80	90
64	89	90	81	56	63	82	90	
65	88	70	45	52	81	91	90	
66	80	54	51	68	90	95	88	
67	81	73	78	91	102	99	106	
68	81	92	98	99	103	109	102	
69	95	96	100	109	101	112	103	

Evaluation

Mutual information - it is a measure of co-occurrence of intensity between the two images



		Moving image		
Fixed image				
		0	0	200
		0	300	0
		100	0	0

Evaluation

Increasing mutual information value

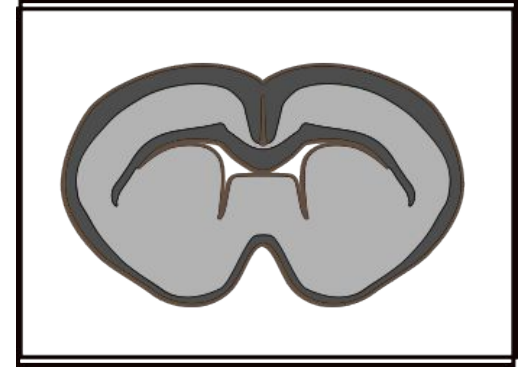
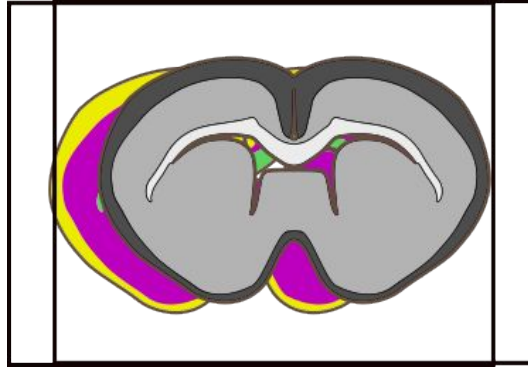
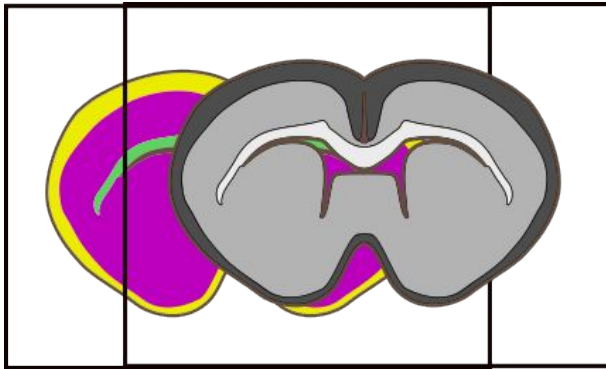
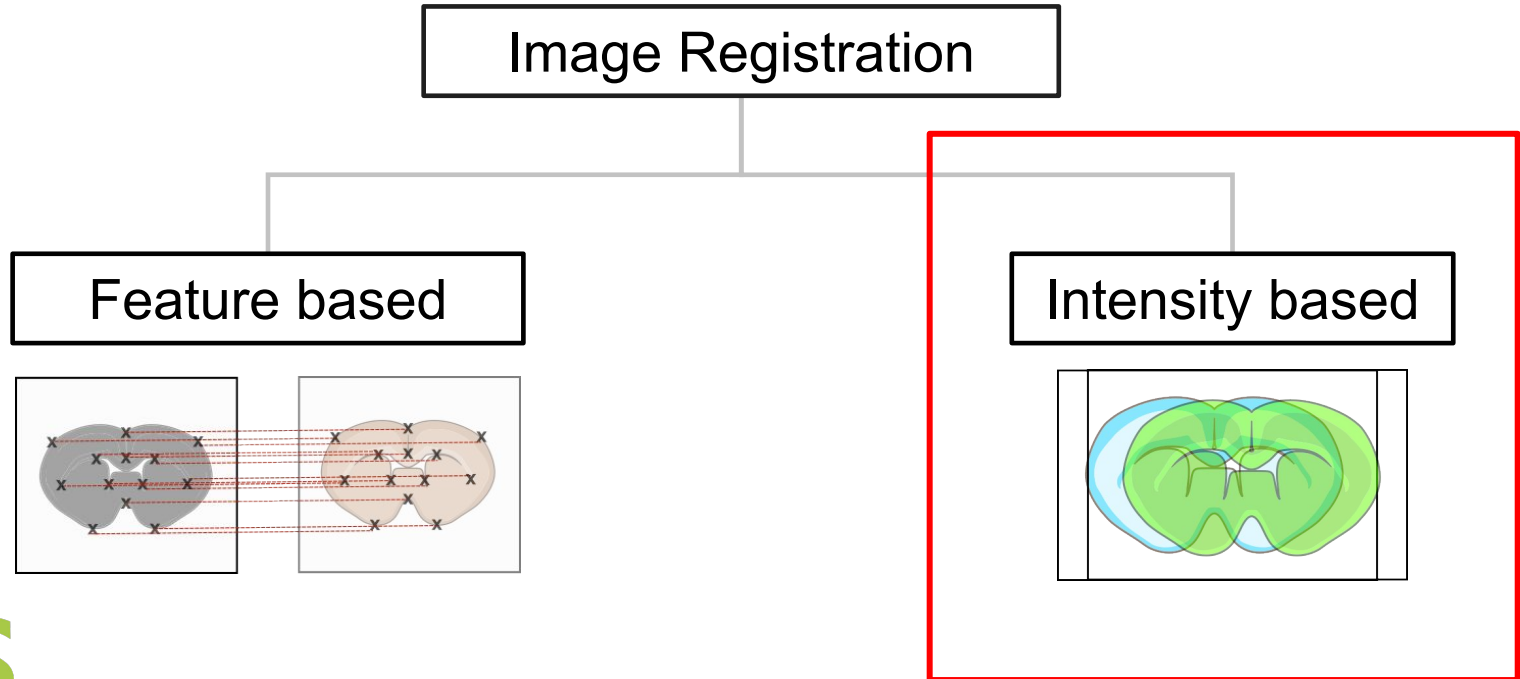
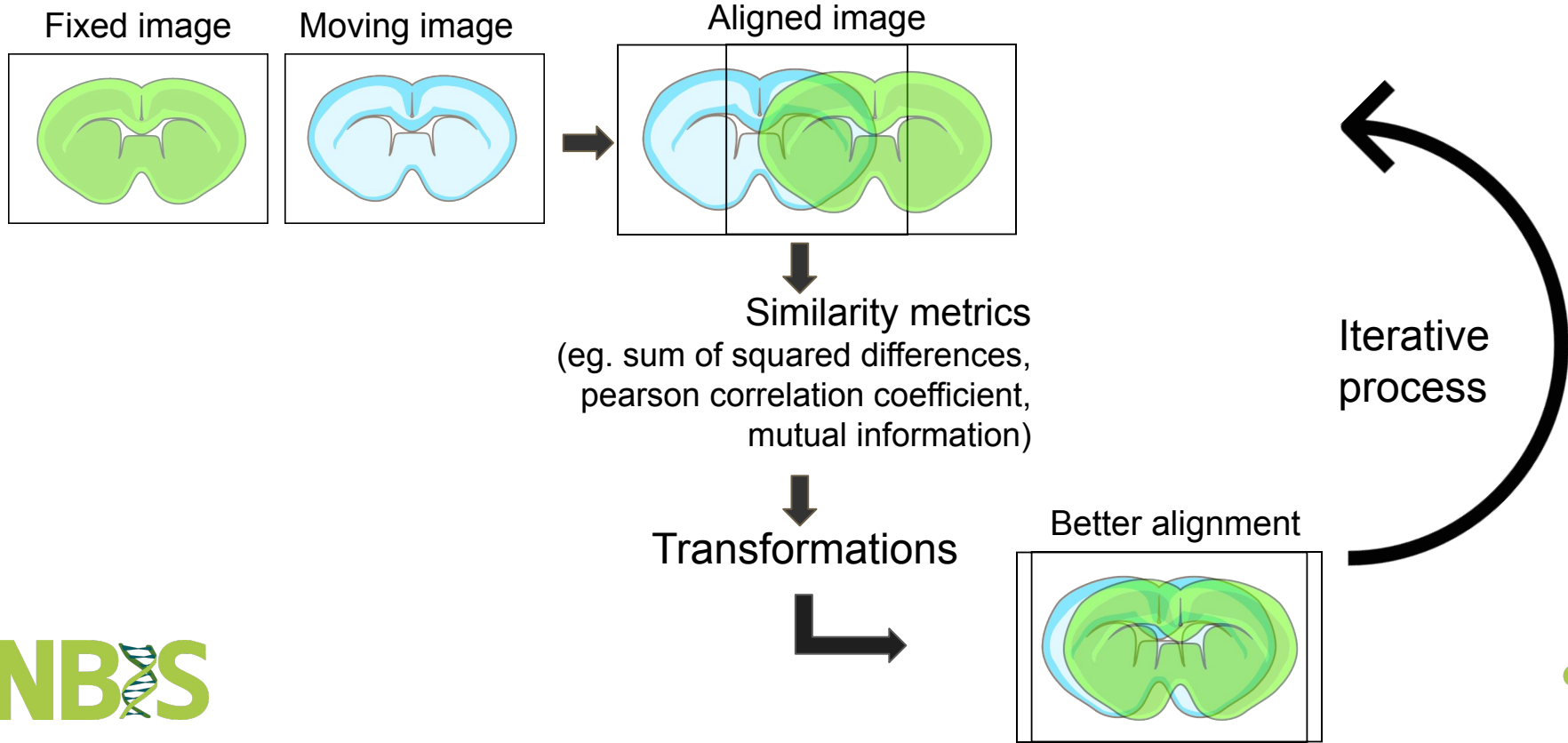


Image registration - Methods

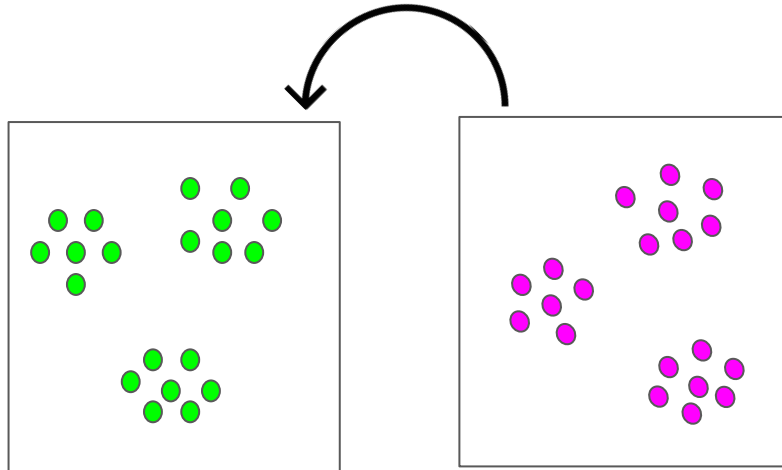


Intensity-based registration



Point set registration

- Assigning correspondence between two sets of points and to determine the transformation that maps one point set to the other
- Methods - Coherent Point Drift (CPD) algorithm, Iterative Closest Point (ICP) and many more



Tools

Tool	Usage	Method of transformation
Warpy	Manual	Used for more complex transformations than affine. Uses ImLib2 library, elastix and BigWarp
BigWarp	Manual	Landmark-based, Transformations - rigid, affine, similarity, thin plate spline
Wsireg	Semi-automated	Uses elastix for transformation Supports linear and non-linear transformation models
Fast4DReg	Automated	A Fiji macro. Mostly used for drift correction but can also be used for aligning multi-channel 2D or 3D images
MultiStackReg	Automated	A Fiji plugin; Based on TurboReg. Mostly for aligning stacks
VALIS	Automated	Rigid and non-rigid transformations
ANTsPy	Automated	Rigid and deformable transformations

Current developments

- [ITK-Wasm](#) - image registrations and spatial analysis can be carried out in a web browser
- Information about the transformations and the coordinate systems can be made available (soon) in the OME-Zarr format. For more details, check out [here](#)

Resources

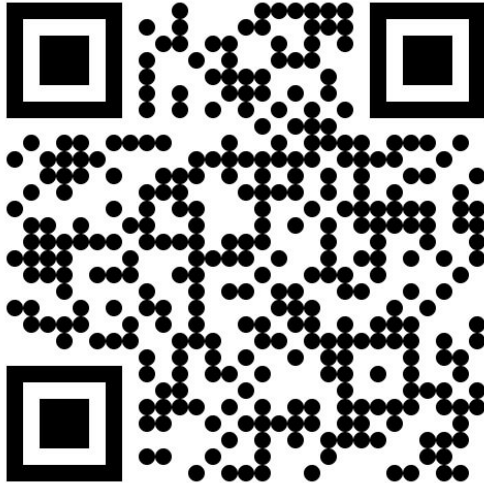
- ITK-Elastix - [GitHub](#)
- Elastix [documentation](#)
- For more image registration tools - [BioImage Informatics Index](#)
- If you have any questions, feel free to post it on [image sc forum](#)

References

- Jeremy L Muhlich, Yu-An Chen, Clarence Yapp, Douglas Russell, Sandro Santagata, Peter K Sorger, Stitching and registering highly multiplexed whole-slide images of tissues and tumors using ASHLAR, *Bioinformatics*, Volume 38, Issue 19, October 2022, Pages 4613–4621, <https://doi.org/10.1093/bioinformatics/btac544>
- Barbara Zitova, Jan Flusser, Image registration methods: a survey. *Volume 21, Issue 11*, October 2003, Pages 977-1000
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- Ntatsis, K., Dekker, N., Valk, V., Birdsong, T., Zukić, D., Klein, S., Staring, M. & McCormick, M. (2023). itk-elastix: Medical image registration in Python. Scipy 2023, <https://doi.org/10.25080/gerudo-f2bc6f59-00d>
- Szeliski, R. (2022). *Computer Vision: Algorithms and Applications*, 2nd ed. Springer.
- Blampey, Q., Mulder, K., Gardet, M. et al. Sopa: a technology-invariant pipeline for analyses of image-based spatial omics. *Nat Commun* 15, 4981 (2024). <https://doi.org/10.1038/s41467-024-48981-z>
- All the mouse brain images used for illustration were from NIAID Visual & Medical Arts. (10/7/2024). Mouse Brain Coronal. NIAID NIH BIOART Source. bioart.niaid.nih.gov/bioart/370
- Some images used for illustration (as indicated) come from the Broad Bioimage Benchmark Collection specifically the dataset from Gustafsdottir et al., 2013
- Images that have no indications are from Bhuvanasundar Renganathan, Research Associate, Northwestern University, Chicago

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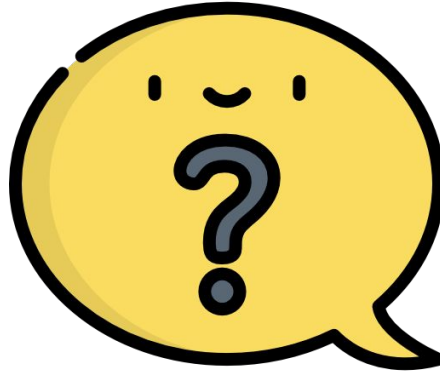


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Questions



Demonstration

DEMO

